

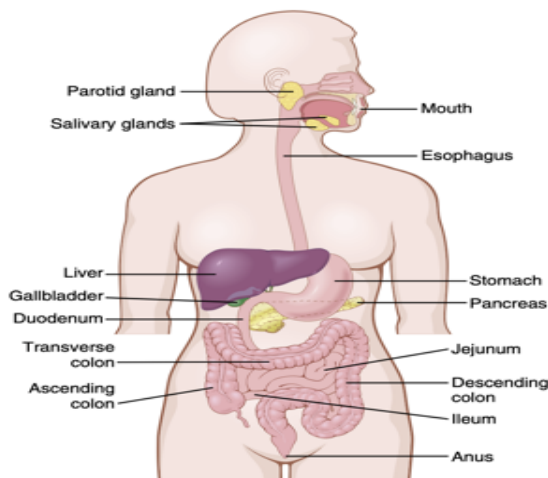
GENERAL PRINCIPLES OF GASTROINTESTINAL FUNCTION

GENERAL PRINCIPLES AND ANATOMY

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OPHT 1-1

- To supply the body with water, electrolytes, vitamins, and nutrients.
- Requires food movement, digestion, absorption, blood circulation, and control by nervous and hormonal systems.
- **5 Layers of Gastrointestinal Wall from the outside:**
 1. Serosa
 2. Longitudinal smooth muscle layer
 3. Circular smooth muscle layer
 4. Submucosa
 5. Mucosa



GASTROINTESTINAL MOTILITY & ELECTRICAL ACTIVITY

GASTROINTESTINAL SYNCYTIUM

- Smooth muscle fibers are connected via gap junctions.
- Gap junctions allow electrical signals to travel easily from cell to cell, causing the muscle layer to function as a syncytium.
- When one part is excited, the action potential travels in ALL directions.

ELECTRICAL WAVES

2 types of electrical waves:

1. **SLOW WAVES** - not action potentials—they are slow, undulating changes in the resting membrane potential.
 - Generated by **interstitial cells of Cajal**.
 - **CAJAL** - electrical pacemakers.
 - **DO NOT** cause muscle contraction, they control the rhythm and appearance of spike potentials.
2. **SPIKE POTENTIALS** - true action potentials that occur when the RMP becomes **more positive than -40mV**.
 - Causes muscle contraction.
 - Appear on the peaks of the slow waves; **higher slow wave peak = greater frequency of spike potentials**.

ROLE OF CALCIUM - caused by the **slow entry of large numbers of calcium ions (and some sodium ions)** through calcium-sodium channels.

- Entry of calcium during spike potentials that directly causes the muscle fibers to contract.

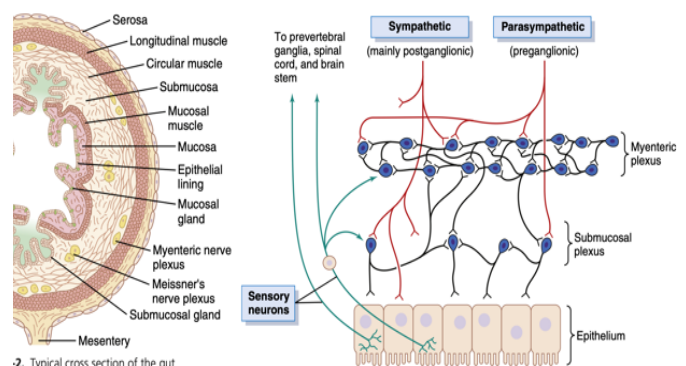
MEMBRANE POTENTIAL CHANGES:

- **DEPOLARIZATION** - makes the muscle **MORE excitable (less negative)**.
 - Caused by:
 - Muscle stretching
 - Acetylcholine (Parasympathetic)
 - GI Hormones
- **HYPERPOLARIZATION** - makes the muscle **LESS excitable (more negative)**.
 - Caused by:
 - Norepinephrine/Epinephrine
 - Sympathetic stimulation

NEURAL CONTROL OF THE GI TRACT

ENTERIC NERVOUS SYSTEM (ENS)

- GI Tract's intrinsic nervous system.
- Essential for **controlling GI movement and secretion**.
- **2 Main Plexuses:**
 1. **MYENTERIC PLEXUS (AUERBACH'S)** - between muscle layers that mainly controls GI movements.
 2. **SUBMUCOSAL PLEXUS (MEISSNER'S)** - located in the submucosa that mainly controls GI secretion and local blood flow.



AUTONOMIC (EXTRINSIC) CONTROL:

- **PARASYMPATHETIC STIMULATION**
 - Travels via **Vagus nerves** (for upper GI tract) and **Sacral nerves** (for distal large intestine).
 - Increases the activity of the entire ENS, enhancing most GI functions.
 - **ACETYLCHOLINE: primary excitatory neurotransmitter**.
- **SYMPATHETIC STIMULATION**
 - Originated from spinal segments T5-L2.
 - Inhibits GI tract activity mainly by the inhibitory effect of Norepinephrine.

GASTROINTESTINAL REFLEXES:

3 Types:

1. **Entirely within the ENS** - for local control of secretion, peristalsis, etc.
2. **Gut to sympathetic ganglia and back** - for long distance signals (gastrocolic reflex).
3. **Gut to the spinal cord/brain stem and back** (vagal, pain, and defecation reflexes).

HORMONAL CONTROL OF MOTILITY

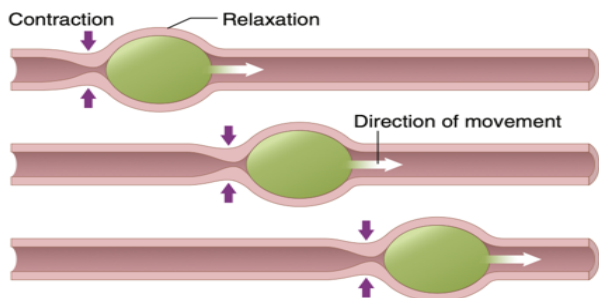
- **GASTRIN** - secreted by **G cells in the stomach**.
 - Stimulates gastric acid secretion and mucosal growth.
- **CHOLECYSTOKININ (CCK)** - secreted by **I cells in the duodenum/jejunum** in response to fat.
 - Strongly contracts the gallbladder; moderately inhibits stomach contraction.
- **SECRETIN** - secreted by **S cells in the duodenum** in response to acid.
 - Promotes pancreatic bicarbonate secretion.
- **GIP (Glucose-dependent Insulinotropic Peptide)** - secreted by the **upper small intestine**.
 - Mildly decreases stomach motor cavity and stimulates insulin secretion.
- **MOTILIN** - secreted **during fasting**.
 - Increases GI motility and stimulates the interdigestive myoelectric complexes (motility waves that occur every 90mins in a fasted state).

FUNCTIONAL MOVEMENTS

1. PROPULSIVE MOVEMENTS (PERISTALSIS)

- Basic propulsive movement, **where a contractile ring forms and moves forward**.
- The usual stimulus is distention of the gut.
- **Active myenteric plexus = effective peristalsis**.
- **“LAW OF THE GUT”** - the myenteric reflex where a **contractile ring forms BEHIND (oral)** the distended segment, and the gut **RELAXES downstream (anal)**, pushing food toward the anus.

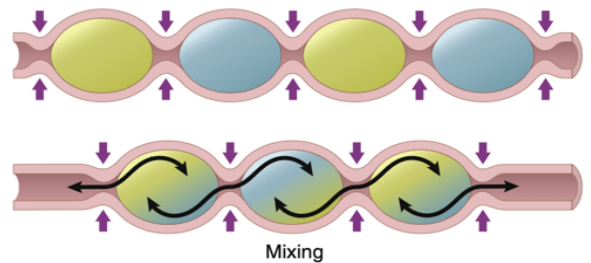
A. Peristalsis contractions (esophagus, stomach, small intestine)



2. MIXING MOVEMENTS (SEGMENTATION)

- **Keeps intestinal contents mixed.**
- Achieved by peristaltic contractions blocked by a sphincter/by local, intermittent segmentation contractions that “CHOP” the contents.

B. Segmentation contractions (small and large intestines)



PROPULSION & MIXING FOOD IN THE ALIMENTARY TRACT

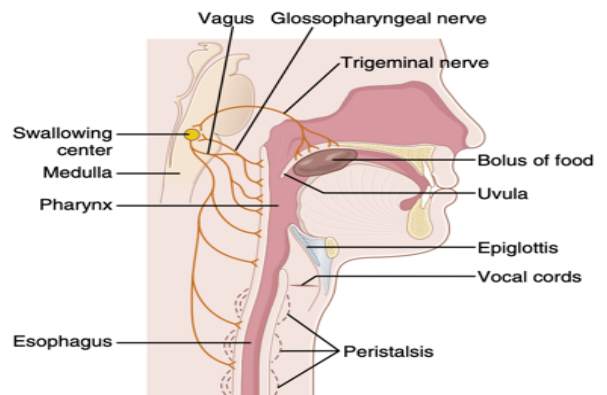
INGESTION AND SWALLOWING (DEGLUTITION)

- **MASTICATION (CHEWING)** - important to break down indigestible cellulose membranes in fruits and vegetables.
 - Controlled by a chewing reflex initiated by food in the mouth.

• SWALLOWING (DEGLUTITION)

3 Stages:

1. **VOLUNTARY STAGE** - the tongue consciously pushes the food bolus to the back of the pharynx.



2. **PHARYNGEAL STAGE (INVOLUNTARY)** - the **MOST CRITICAL** stage and is an **AUTONOMIC** reflex.

- Triggered when food stimulates receptors (especially on tonsillar pillars).

- **AIRWAY PROTECTION:** soft palate closes the nasal passages; vocal cords are pulled tightly together; epiglottis swings back to cover the larynx.
- **PROPULSION:** Upper Esophageal Sphincter (UES) relaxes and a fast peristaltic wave forces food into the esophagus.
- **MEDULLA AND PONS:** the swallowing center in the brain stem that controls this entire system.

3. **ESOPHAGEAL STAGE (INVOLUNTARY)** - transports food to the stomach.

- **PRIMARY PERISTALSIS:** continuation of swallowing waves from the pharynx. (8-10secs to reach the stomach).
- **SECONDARY PERISTALSIS:** food remains in the esophagus. (distention

triggers these backup waves to clear the passage.

- **RECEPTIVE RELAXATION:** as the peristaltic wave travels, the Lower Esophageal Sphincter (LES) and the upper part of the stomach relax AHEAD of the food.
- **Lower Esophageal Sphincter (LES):** remains tonically to prevent acidic gastric reflux into the esophagus.

MOTOR FUNCTIONS

1. STORAGE

- Food is **stored in the body and fundus of the stomach.**
- A **vagovagal reflex** (from the stomach to the brain stem and back) **reduces muscle tone in the stomach wall, allowing it to relax** and accommodate **0.8-1.5L of food** with little increase in pressure.

2. MIXING AND PROPULSION (To form Chyme)

- Weak mixing waves **begin in the midstomach** and **move toward the antrum every 15-20 seconds.**
- Weak mixing waves create **high-pressure constrictor rings.**
- **RETROPULSION** - the pyloric opening is small, so as the wave hits the near-closed pylorus, most of the antral contents are squeezed backward (retropulsion).
 - ◆ This is the **most important mixing mechanism in the stomach.**
- **PYLORIC PUMP** - intense peristaltic contractions in the antrum are responsible for "pumping" small amounts of chyme into the duodenum.

- 3. **Hunger Contractions** - intense rhythmic contractions that can fuse into a sustained contraction (**2-3mins**) when the stomach has been empty for many hours.

REGULATION OF STOMACH EMPTYING

The rate of emptying is precisely controlled, to ensure the small intestine is not overwhelmed.

- **PROMOTING FACTORS (from the Stomach):**
 - **GASTRIC FOOD VOLUME** - Stretching the stomach wall initiates **myenteric reflexes** that **enhance the pyloric pump.**
 - **HORMONE (GASTRIN)** - mildly enhances the activity of the pyloric pump.
- **INHIBITORY FACTORS (from the Duodenum)**
 - Dominant control factors.

1. ENTEROGASTRIC NERVOUS REFLEXES

- Triggered by duodenal distention, irritation, acidity (pH below 3.5-4), or chyme that is **hypertonic/hypotonic** or **contains protein/fat breakdown products.**
- **Inhibit pyloric pump and increase the tone of the pyloric sphincter,** significantly slowing emptying.

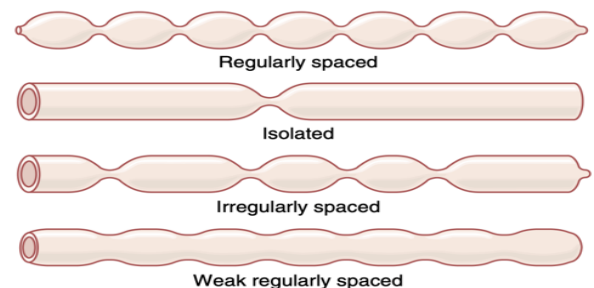
2. HORMONAL FEEDBACK

- Presence of fat in the duodenum = **most potent stimulus for inhibitory hormones.**
- **CHOLECYSTOKININ (CCK)- MOST POTENT HORMONE** released in response to fat.
 - Directly inhibits the stomach motility caused by gastrin.
- **SECRETIN** (released due to acid) and **GIP** - Minor roles in inhibiting emptying

MOVEMENTS OF THE SMALL INTESTINE

MIXING CONTRACTIONS (SEGMENTATION)

- **Primary mixing movements**, distinct from peristalsis.
- Concentric contractions that "chop" the chyme divides the intestine into segments resembling a chain of sausages.

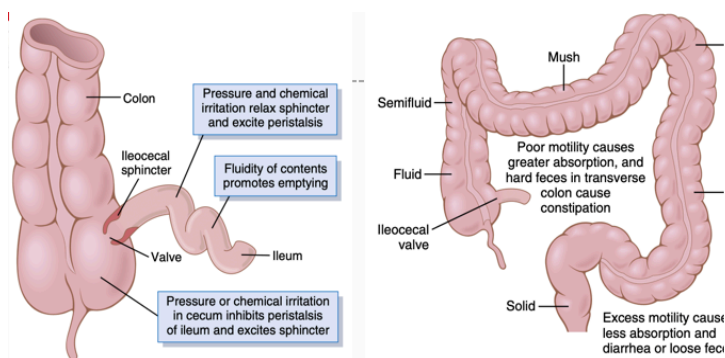


PROPULSIVE MOVEMENTS (PERISTALSIS)

- **Weak, propulsive waves** that typically die out after **3-5 cm.**
- **Net movement is very slow (about 1 cm/min),** meaning chyme takes **3 to 5 hours to travel from the pylorus to the colon.**
- **GASTROILEAL REFLEX** - a reflex that intensifies peristalsis in the ileum, **pushing chyme into the large intestine.**
- **PERISTALTIC RUSH** - abnormal, powerful, and rapid wave of peristalsis caused by intense irritation to **quickly sweep contents into the colon.**

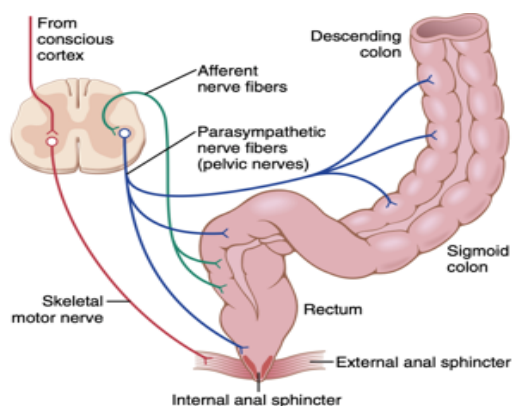
MOVEMENTS OF THE COLON & DEFECATION

- **PROXIMAL COLON** - for absorption of water and electrolytes.
- **DISTAL COLON** - for storage of feces. Movements are normally sluggish.
- **ILEOCECAL VALVE** - protrudes into the cecum, along with the ileocecal sphincter, **prevents the backflow of feces from the colon into the small intestine.**



- **MIXING MOVEMENTS (HAUSTRATIONS)**
 - **"Segmentation" movements of the colon.**
 - Combined contractions of the circular and longitudinal (teniae coli) muscles cause the colon to bulge into bag-like sacs called haustrations.
 - This **"digs and rolls"** the fecal material, exposing all of it to the mucosal surface for fluid absorption.
- **PROPULSIVE MOVEMENTS (MASS MOVEMENTS)**
 - Primary propulsive force in the colon, occurring **1-3 times daily (often after breakfast).**
 - A segment of the colon (often 20+ cm) contracts as a unit, propelling the fecal mass down the colon.
 - Triggered by the gastrocolic and duodenocolic reflexes-initiated by stomach and duodenal distention.
- **DEFECATION**
 - Rectum is usually empty. MASS MOVEMENTS push feces into the rectum, triggering the urge to defecate.

- **Anal Sphincters:**
 - **INTERNAL** - involuntary smooth muscle.
 - **EXTERNAL** - voluntary striated muscle (controlled by the pudendal).



DEFECATION REFLEXES

- 1. INTRINSIC REFLEX (WEAK)**
 - Rectal distention **activates the myenteric plexus**, causing weak peristalsis and relaxing the internal sphincter.
- 2. PARASYMPATHETIC REFLEX (STRONG)**
 - Rectal distention also **sends signals to the sacral spinal cord.**

- The cord sends signals back via parasympathetic nerves that greatly intensify peristalsis and relax the internal sphincter.

Defecation occurs **only when the external anal sphincter is also voluntarily relaxed.**

SECRETORY FUNCTIONS OF ALIMENTARY TRACT

GENERAL PRINCIPLE OF SECRETION

- **PRIMARY FUNCTIONS:**
 1. Secreting digestive enzymes.
 2. Secreting mucus for lubrication and protection.

TYPES OF GLANDS:

- 1. MUCOUS (GOBLET) CELLS** - Single cells on the gut surface that **secrete mucus in response to local irritation.**
- 2. PITS (CRYPTS OF LIEBERKUHN)**
 - Invaginations in the epithelium, found in the small intestine that **contain specialized secretory cells.**
- 3. TUBULAR GLANDS** - Deep glands found in the stomach and upper duodenum (oxyntic glands).
- 4. COMPLEX GLANDS** - The salivary glands, pancreas, and liver, which are **located outside the alimentary tract wall and empty into it via ducts.**

BASIC STIMULATION OF GLANDS

ENTERIC NERVOUS SYSTEM

- Presence of food (via tactile stimulation, chemical irritation, or distention) activates the local enteric nervous system, which in turn stimulates gland secretion.

PARASYMPATHETIC STIMULATION

- Almost **invariably increases secretion rates**, especially for the salivary glands, gastric glands, and pancreas.

SYMPATHETIC STIMULATION

Dual effect:

1. It can **slightly increase secretion on its own**, but,
2. If secretion is already high (due to parasympathetic or hormonal drive), **sympathetic stimulation reduces secretion by constricting blood vessels to the glands.**

HORMONES

- **Gastrin** is released from the GI mucosa into the blood and travels to glands **to stimulate secretion.**

SALIVARY SECRETION

GLANDS AND SECRETION TYPES:

- **Parotid** (secrete almost entirely serous fluid), **submandibular**, and **sublingual** (secrete a mix).

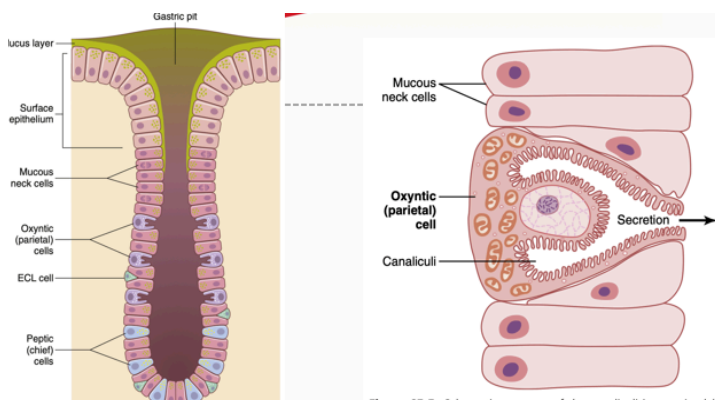
- **2 types of protein secretion:**
 1. **Serous secretion** with ptyalin (an alpha-amylase) for digesting starches.
 2. **Mucus secretion** with mucin for lubrication.

ESOPHAGEAL SECRETION

- Entirely **mucous**.
- Provide **lubrication** for **swallowing**.
- Compound glands near the stomach junction secrete mucus to protect the esophagus from acidic gastric reflux.

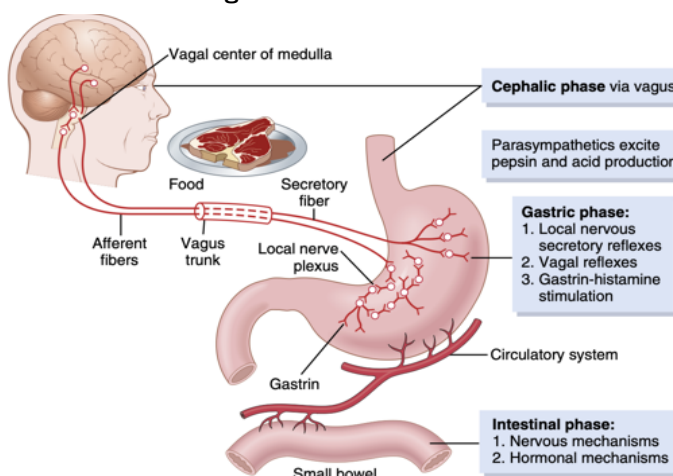
KEY CELL TYPES OF OXYNTIC GLANDS

- **MUCOUS NECK CELLS** - secrete **mucus**.
- **PEPTIC (CHIEF) CELLS** - secrete **pepsinogen**.
- **PARIETAL (OXYNTIC) CELLS** - secrete **hydrochloric acid (HCl)** and **intrinsic factor**, essential for the **absorption of vitamin B12** in the **ileum**.
- **ENTEROCHROMAFFIN-LIKE (ECL) CELLS** - secrete **histamine**.



PHASES OF GASTRIC SECRETION

1. **CEPHALIC PHASE (30%)** - before food enters the stomach.
 - Triggered by the **sight, smell, or thought of food**, and the signal is transmitted by the vagus nerve.
2. **GASTRIC PHASE (60%)** - when food enters the stomach.
 - Triggers **long vagovagal reflexes**, **local enteric reflexes**, and the **gastrin-histamine mechanism**.
3. **INTESTINAL PHASE (10%)** - presence of food in the duodenum causes a small amount of continued gastric secretion.



PANCREATIC SECRETION

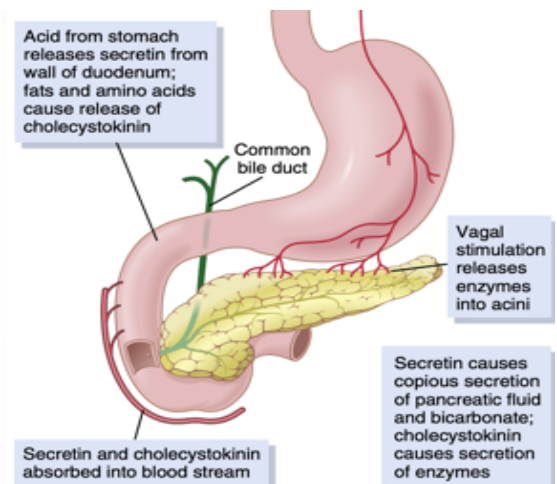
PANCREAS - secretes digestive enzymes from its acini and a large volume of sodium bicarbonate solution from its ducts.

PANCREATIC DIGESTIVE ENZYMES:

1. **PROTEIN** - Trypsin, chymotrypsin, and carboxypolypeptidase.
2. **CARBOHYDRATE** - Pancreatic amylase (digests starch).
3. **FAT** - Pancreatic lipase, cholesterol esterase, and phospholipase.

REGULATION OF PANCREATIC SECRETION

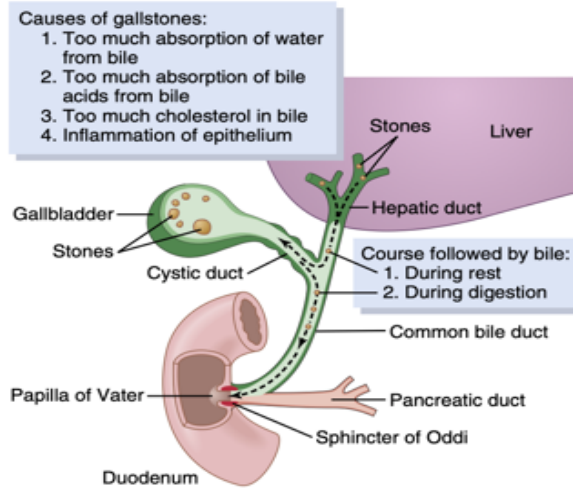
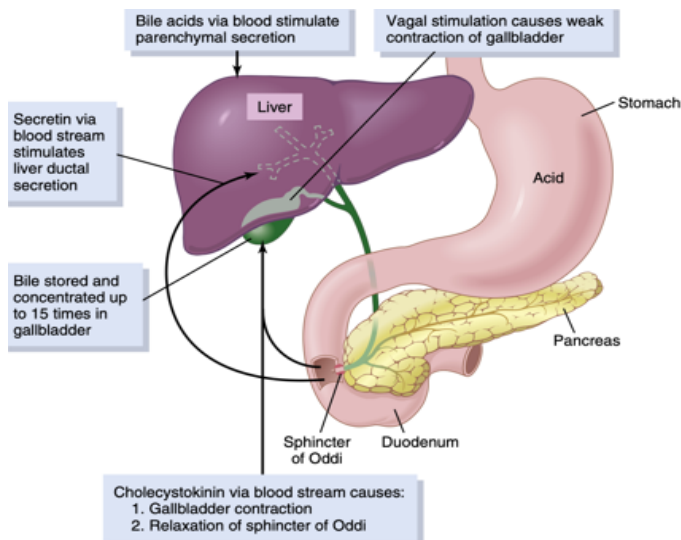
- **ACETYLCHOLINE (from Vagus)** - stimulates the acinar cells to **secrete digestive enzymes**.
- **CHOLECYSTOKININ (CCK)** - secreted by "**I cells**" in the **duodenum** in response to fats and proteins (**peptones**).
 - Via blood and strongly stimulates the acinar cells to **secrete large amounts of digestive enzymes**.
- **SECRETIN** - secreted by "**S cells**" in the duodenum in response to acid chyme (pH < 4.5).
 - Via blood and strongly stimulates the ductal cells to **secrete large volumes of sodium bicarbonate and water**.



BILE SECRETION (LIVER) & GALLBLADDER FUNCTION

FUNCTIONS OF BILE:

1. **FAT DIGESTION/ABSORPTION** - Bile acids emulsify (break down) large fat globules and form micelles, which are essential for "ferrying" digested fats to the intestinal mucosa for absorption.
2. **EXCRETION** - Bile is the **primary route for excreting waste products** like bilirubin (from hemoglobin destruction) and excess cholesterol.



3. SECRETION AND STORAGE

- **Bile is secreted in 2 Stages:**
 1. **HEPATOCYTES** - secrete the primary bile (bile acids, cholesterol).
 2. **DUCTS** - add a watery, bicarbonate-rich solution (stimulated by secretin).
- The gallbladder concentrates bile 5- to 20-fold by actively absorbing water, Na⁺, Cl⁻, and other small electrolytes.

EMPTYING OF THE GALLBLADDER

- **CCK** - the **most potent stimulus for gallbladder emptying**.
 - Released **from the duodenum** when fatty foods enter.
 - **Causes 2 simultaneous events:**
 1. **Contraction of the gallbladder.**
 2. **Relaxation of the sphincter of Oddi** (allows bile to flow into the duodenum).

ENTEROHEPATIC CIRCULATION OF BILE SALTS

- **94% of bile salts** are reabsorbed from the small intestine (especially the **distal ileum**) back into the portal blood.
- **LIVER** - re-absorbs these salts and re-secretes them.
- This recirculation is crucial, as the body's small pool of bile salts cycles **~17 times per day** to produce enough bile.

GALLSTONES

- **BILE SALTS AND LECITHIN** - keep cholesterol (which is insoluble in water) dissolved in micelles.
- Under abnormal conditions (high-fat diet, gallbladder inflammation), cholesterol may precipitate out of solution, forming cholesterol gallstones.

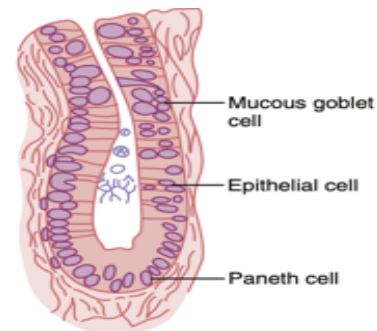
SECRETIONS OF THE SMALL INTESTINE

BRUNNER'S GLANDS (DUODENUM)

- Secrete **large amounts of alkaline mucus**.
- Protect the duodenal wall from the highly acidic gastric juice.
- Inhibited by sympathetic stimulation, which **may be a factor in duodenal ulcers**.

CRYPTS OF LIEBERKUHN (ENTIRE SMALL INTESTINE)

- Enterocytes in the crypts **secrete a large volume of watery fluid** (alkaline, extracellular fluid-like solution).
- **"watery vehicle"** to help absorb nutrients from the chyme.



DIGESTIVE ENZYMES (SMALL INTERNAL INTESTINE)

- Enzymes are located **within the enterocytes that cover the villi**.
- Enzymes digest food as it is being absorbed and include:
 1. **Peptidases**
 2. **Disaccharidases** (sucrase, maltase, lactase)
 3. **Intestinal lipase**

SECRETIONS OF THE LARGE INTESTINE

LARGE INTESTINE MUCOSA - has many crypts of Lieberkuhn but **contains no villi**.

- Epithelial cells **secrete almost no digestive enzymes**.
- The primary secretion: **mucus from numerous mucous cells**.

Function of Mucus:

- Protects the intestinal wall from **excoriation and bacterial activity**.
- Provides an **"adherent medium"** for holding fecal matter together.

- Contains **bicarbonate**, providing an alkaline (pH 8.0) barrier against acids formed by bacteria in the feces.

DIGESTION & ABSORPTION IN THE GASTROINTESTINAL TRACT

DIGESTION OF CARBOHYDRATES

PRIMARY SOURCES

- The main dietary carbohydrates are **sucrose** (table sugar), **lactose** (milk sugar), and **starches** (polysaccharides).

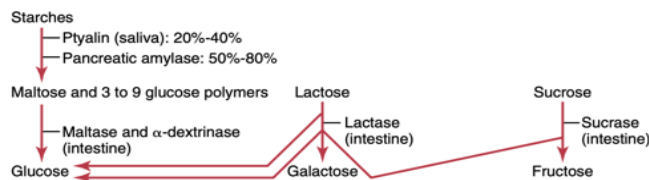
IN THE MOUTH & STOMACH

- Digestion begins in the mouth with the enzyme **ptyalin** (salivary alpha-amylase), breaking down starches (continues in the stomach for up to an hour until stomach acid (pH < 4.0) inactivates the enzyme).

IN THE SMALL INTESTINE (PANCREATIC)

- **Pancreatic alpha-amylase** - breaks down the remaining starches into maltose and other small glucose polymers.

AT THE BRUSH BORDER - final breakdown.



Enzymes located in intestinal cells split disaccharides into monosaccharides:

- **LACTASE** splits lactose glucose + galactose
- **SUCRASE** splits sucrose glucose + fructose
- **MALTASE AND ALPHA-DEXTRINASE** split maltose/glucose polymers glucose

FINAL PRODUCTS - monosaccharides (glucose, galactose, and fructose).

- **Glucose** accounts for **over 80% of the final product**.

DIGESTION OF PROTEINS

IN THE STOMACH

- **PEPSIN** - begins protein digestion.
 - **Important for digesting collagen** (connective tissue in meats).
 - Breaks proteins down into **smaller polypeptides** (10-20% of total digestion).

IN THE SMALL INTESTINE (PANCREATIC)

- Most protein digestion occurs in the **duodenum and jejunum**.
- **PANCREATIC ENZYMES** (trypsin, chymotrypsin, carboxypolypeptidase, and elastase) **break large polypeptides into smaller tripeptides and dipeptides**.

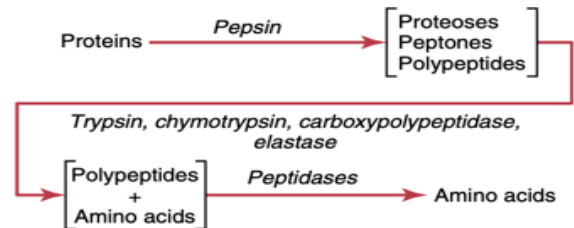
AT THE BRUSH BORDER

- Peptidases in the enterocyte membrane **split the peptides into dipeptides, tripeptides, and a few single amino acids**.

INSIDE THE CELL

- **DIPEPTIDES AND TRIPEPTIDES** - transported into the intestinal cell, where intracellular peptidases split them into their final form.

FINAL PRODUCT - over 99% of protein is absorbed as individual amino acids.



DIGESTION OF FATS

PRIMARY SOURCE

- **TRIGLYCERIDES** (Neutral fats) - most abundant dietary fats

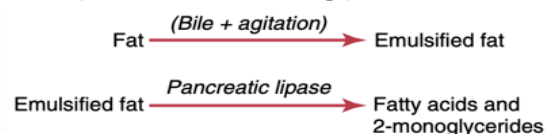
STEP 1: EMULSIFICATION (in the Duodenum):

- **Critical first step.**
- **Large fat globules are broken down into tiny droplets** by agitation.
- Made possible by bile salts and **LECITHIN** (from liver bile), which **act as detergents**.
- Emulsification massively increases the surface area for enzymes to work on.

STEP 2: HYDROLYSIS (in the Small Intestine):

- **PANCREATIC LIPASE** - most important fat-digesting enzyme
 - Attacks the surface of the emulsified droplets and hydrolyzes the triglycerides.

FINAL PRODUCTS - the end products of fat digestion are free fatty acids and 2-monoglycerides.



STEP 3: MICELLE FORMATION

- **MICELLES** - small spherical globules that have a fat-soluble center and a water-soluble exterior formed by bile salts.
 - Traps the fatty acids and monoglycerides.
- **"FERRYING" FUNCTION**
 - Micelles are essential because they "ferry" these insoluble fat products through the watery chyme to the surface of the intestinal cells, where they can be absorbed.

ABSORPTION OF NUTRIENTS (SMALL INTESTINE)

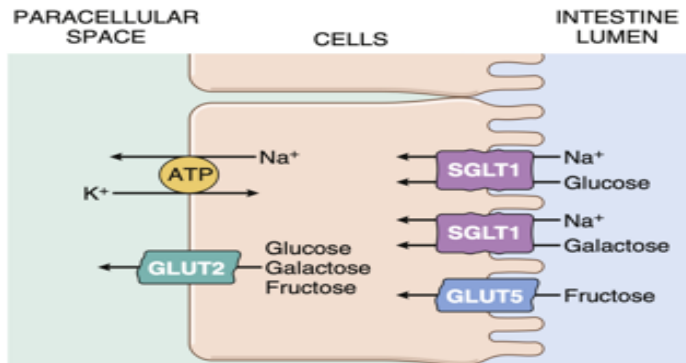
CARBOHYDRATES (MONOSACCHARIDES)

- **Glucose and Galactose:**
 - Absorbed **via secondary active transport**, powered by the sodium gradient.

- The SGLT1 protein co-transporters one Na⁺ with one glucose/galactose molecule into the cell.

- **Fructose:**

- Absorbed by facilitated diffusion via the **GLUT5 transporter**.
- **Does not require sodium and is slower.**
- All three monosaccharides exit the cell into the blood via **GLUT2 transporter**.



- **Proteins (Amino Acids):**

- Absorbed (as dipeptides, tripeptides, or amino acids) via **sodium co-transport (secondary active transport), similar to glucose**.
- Any peptides that enter the cell are **broken into single amino acids before entering the blood**.

- **Fats** - Micelles bring fatty acids and monoglycerides to the cell surface, where they diffuse directly across the cell membrane (because they are **lipid-soluble**).

(INSIDE THE CELL)

- **FATTY ACIDS AND MONOGLYCERIDES** - taken to the smooth endoplasmic reticulum and re-synthesized back into triglycerides.
- **CHYLOMICRON FORMATION** - new packaged triglycerides with cholesterol and proteins into large globules called **chylomicrons**.
- **ENTRY INTO CIRCULATION**
 - Chylomicrons are too large for capillaries. They are released from the cell base and enter the lymphatic system via the central lacteal of the villus, bypassing the liver.

EXCEPTION - Short- and medium-chain fatty acids (from butterfat) are water-soluble and are absorbed directly into the portal blood, not the lymph.

ABSORPTION IN THE LARGE INTESTINE

- **PROXIMAL HALF (absorbing colon)** - absorbs water and electrolytes
- **DISTAL HALF (storage colon)** stores feces.

ABSORPTION

- Large intestine **absorbs Na⁺ actively and water passively (via osmosis)**.

- **Its tight junctions are much tighter than the small intestine's**, allowing it to absorb Na⁺ against a very high concentration gradient and preventing ion back-leak.

BACTERIAL COLON

- **Colon bacilli (bacteria)** are present and perform several functions:
 - **Digest small amounts of cellulose** (negligible energy for humans).
 - **Produce vitamin K** (essential for blood coagulation, often insufficient in the diet).
 - **Produce vitamin B12, thiamine, and flatus gases**.

FECES

- Normally **3/4 water and 1/4 solid**.
- **SOLID WATER** - consists of **dead bacteria, undigested roughage, fat, and inorganic matter**.
- **STERCOBILIN AND UROBILIN (BILIRUBIN DERIVATIVES)** - where the brown color came from.